MECH 191 — Metallurgy

Week 1

Atomic Structure & Bonding

August 24, 2026 · 5:00 – 7:30 PM · Kai Santos

Week 1 — At a Glance

Topics

- What is metallurgy — and why does it matter?
- The atom: protons, neutrons, electrons, valence shells
- The Periodic Table — groups, periods, and metals
- Atomic bonding — metallic, ionic, covalent
- Alloys and solid solutions
- Crystal structures — SC, BCC, FCC, HCP
- Grains and grain boundaries

Resources

Reference Material

Moniz 5th — Ch. 4
Structure of Metals

Kalpakjian 6th — Ch. 3
Structure and Manufacturing Properties of Metals

Visualizers

- Crystal Structures Visualization
- Periodic Table of Elements
- Grain Boundaries and Types

Metallography Lab — Module 1

EXAMET-5-R-600-HD · 50–500×

Module 1 · Week 1

Atomic Structure & Bonding

What to do this week:

Observe grain boundaries at 100×. Sketch the grain network and estimate average grain diameter.

Steps:

1 Observe — complete feature table
2 Sketch — label magnification & specimen
3 Answer — 4 short questions

Specimens to Check Out

· Specimen A — 1018 low-carbon steel

How to check out the kit:

1 See instructor after class today
2 Sign out the specimen tray
3 Use the EXAMET-5 at your convenience
4 Return kit before leaving class
5 Submit completed module with your portfolio

What is Metallurgy?

The science of metals — understanding structure, properties, and behavior so we can choose, test, treat, and work with them effectively.

The Bronze Example — The First Alloy

Copper: abundant, conductive — but weak. Adding ~10–20% Tin disrupts copper's crystal lattice, creating Bronze: significantly harder and stronger than either metal alone, while keeping copper's workability and corrosion resistance. The first alloy in human history.

Why use alloys?

Pure metals rarely provide the properties we need. Alloys let us engineer the material for the job.

How do alloys work?

Adding a second element disrupts the crystal structure, blocking deformation and changing properties.

What comes next?

Carbon in iron → steel. Chromium in steel → stainless. Every unit in this course builds on this idea.

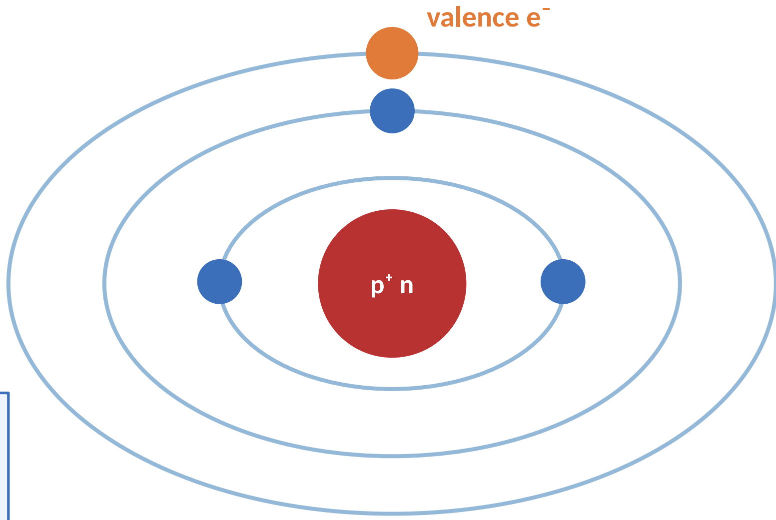
The Atom

Three key components:

Protons
Positive charge — inside the nucleus — defines the element (atomic number)

Neutrons
No charge — inside the nucleus — defines mass. Isotopes vary in neutron count.

Electrons
Negative charge — orbit in energy bands/shells around the nucleus



Valence Electrons — The Bonding Shell

The outermost electrons — the only ones involved in bonding. Carbon has 4 valence electrons, bonding in 4 directions simultaneously. This is why adding carbon to iron creates steel — carbon atoms sit between iron atoms and block deformation.

The Periodic Table

Periods (Rows)

Organized by electron shells. Moving down adds shells — each period is a new shell level.

Groups (Columns)

Organized by valence electrons. Same group = similar bonding behavior and chemical properties.

Our Focus: Groups 3–12

Transition metals — Fe, Cu, Ni, Cr, Ti, Mo, and all engineering metals live here.

Classification by the staircase line:

Metals

Left of the staircase

- Electrically & thermally conductive
- Malleable and ductile
- Lustrous (shiny surface)
- Solid at room temp (except Hg)

Metalloids

The staircase itself

- Properties depend on conditions
- Semiconducting — critical for electronics
- Behavior: part metal, part non-metal
- Si, Ge, As, Sb, Te, Po, At

Non-metals

Right of the staircase

- Poor electrical conductors
- Brittle (if solid) or gaseous
- Non-lustrous
- Tend to gain electrons

Atomic Bonding

The Octet Rule: Atoms are most stable with 8 valence electrons — they bond (lose, gain, or share electrons) to reach that stable state.

Metallic Bond

Metal + Metal

Electrons leave individual atoms and form a shared 'electron sea' — flowing freely through the structure.

Effect on material:

Conducts heat & electricity · Bends without fracturing · Atoms slide past each other

← Our primary focus

Ionic Bond

Metal + Non-metal

One atom gives away electrons; the other takes them. Creates oppositely charged ions that attract strongly.

Effect on material:

High melting point · Hard but brittle · Conducts only when molten or dissolved

Covalent Bond

Non-metal + Non-metal

Neither atom gives up electrons — they share them instead, forming strong directional bonds.

Effect on material:

Extremely strong · Poor conductors · Can be very hard (diamond = covalent network)

Alloys and Solid Solutions

An alloy is an intentional mixture of two or more elements engineered for desired properties — most engineering metals are alloys.

Substitutional Solid Solution

Works when atoms are within ~15% of each other in size

Added atoms swap into lattice positions of the host — like replacing similarly sized bricks. The crystal structure remains but is strained.

Example:

Copper-Nickel (Cu-Ni)
Brass (Cu + Zn)
Many stainless steel alloying additions

Interstitial Solid Solution

Added atoms are much smaller than the host atoms

Small atoms fit into gaps between host atoms — no host atom is displaced. Even tiny amounts dramatically change properties.

Example:

Carbon in Iron → Steel
Carbon atoms sit in gaps between Fe atoms, stiffening the lattice and blocking slip

Crystal Structures — Unit Cells

When metals solidify, atoms arrange into repeating 3D patterns. The unit cell is the smallest repeating building block — like a tile in a tiled floor.

SC

Simple Cubic

Packing: 52%

Ductility: —

Atoms at 8 corners only. Teaching example — not found in real engineering metals.

BCC

Body-Centered Cubic

Packing: 68%

Ductility: Low

Corners + 1 center atom.
Stronger, less ductile.
Fe (room temp), W, Cr, Mo, V

FCC

Face-Centered Cubic

Packing: 74%

Ductility: High

Corners + center + face atoms. Most ductile — most slip planes.
Al, Cu, Ni, Fe (>912°C), Au, Ag

HCP

Hexagonal Close-Packed

Packing: 74%

Ductility: Med

Same packing as FCC, fewer slip planes → less ductile.
Ti, Mg, Zn, Co

Grains and Grain Boundaries

From unit cell to bulk metal:

Unit Cell

Single repeating building block (~0.3 nm). All atoms perfectly aligned.

Crystal / Grain

Millions of unit cells — all with the same orientation. One crystal domain.

Polycrystalline Metal

Countless grains packed together, each pointing a different direction.

Why Grain Size Matters

Grain boundaries block dislocation movement — the mechanism by which deformation spreads through metal. More boundaries = stronger barriers to deformation.

Smaller grains → more boundaries → stronger, tougher material

Grain boundaries = white lines between regions

Key Takeaways

1

Metallic bonding explains metals

Shared electron sea → electrical and thermal conductivity. Non-directional bonding → ductility. This is what separates metals.

2

Navigate the periodic table

Transition metals (Groups 3–12) are the focus. Elements in the same column share behavior — useful for predicting alloy effects.

3

Structure exists at every scale

Atoms arrange into unit cells → unit cells form grains → grains make bulk metal. Each level determines performance.

4

Know BCC, FCC, and HCP

FCC is most ductile (74%, many slip planes). BCC is stronger but less ductile. HCP has 74% packing but fewer slip planes — less ductile than FCC.

Resources & What's Next

Week 1 Resources

- Moniz Ch. 4 — Structure of Metals
- Kalpakjian Ch. 3 — Structure and Manufacturing Properties of Metals

Visualizers

- Crystal Structures Visualization
- Periodic Table of Elements

Coming Up — Week 2

Mechanical Properties & Behavior

- Stress and strain — the language of mechanical behavior
- Elastic vs. plastic deformation and the yield point
- Crystal defects — point, line, planar, and volume types
- Grain boundaries, Hall-Petch, and diffusion through solid metal